Module 7

Capital Budgeting

FINE 3010, Financial Management

Prof. Renping Li, Tulane University

Goals

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- How to decide what project(s) to do? The tool is the NPV rule
- But to use NPV, we need a project's cash flows
- Here, combine what we know about accounting and cash flow effects (**Module 2**) with the NPV rule (**Module 6**), to decide whether or not to accept a project
- **Capital budgeting**, the process of analyzing capital investment opportunities ("projects") and deciding which ones to accept

Roadmap

Should we invest in a project (e.g., a data center)?

NPV > 0?

Recall from **Module 6**

Roadmap

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Should we invest in a project (e.g., a data center)?

How to forecast cash flows?

- Similar to **Module 2**, indirect method for cash flow
- But different, cash flow from a project, not a whole firm

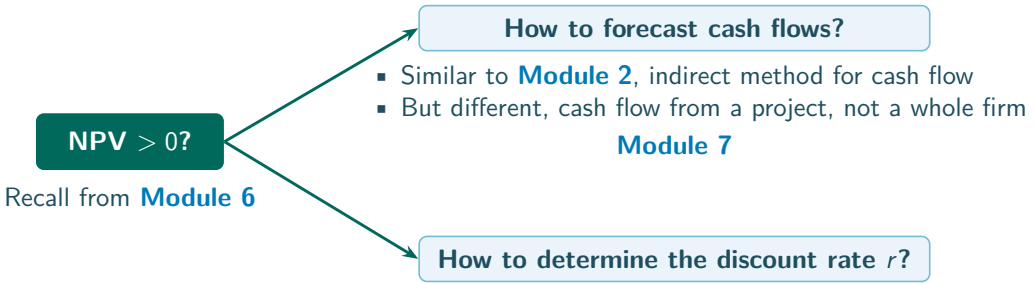
Module 7

NPV > 0?

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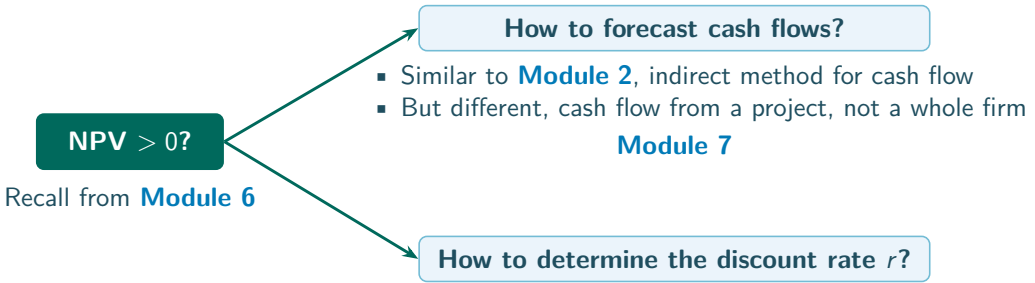
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Motivating Example, Is AI Spending a Bubble?



CNBC, August 12, 2026 (7:21), "We're 'absolutely' in an AI bubble right now, says Lead Edge Capital's Mitchell Green"

Module 7, Part 0

Review of the Statement of Cash Flows

Statement of Cash Flows

Recall from **Module 2**

- **Cash flow**, cash actually received minus cash actually paid, no matter when the sale was made

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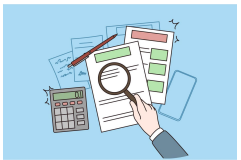
Recall from **Module 2**

Direct method	Indirect method
Lists every cash receipt and payment	Starts from net income and adjusts it
More time-consuming to prepare	Faster to prepare
Rarely used in practice	**The common choice**

Direct →



Indirect →



Statement of Cash Flows

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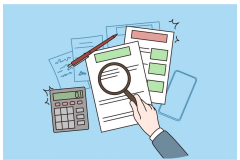
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Indirect →



Statement of Cash Flows

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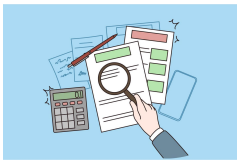
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- Both methods give the same totals
- We will follow the indirect method, starting from net income in the income statement

Direct →



Indirect →



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Effect on net income	0	**−0.4**	**−0.4**	**−0.4**	**−0.4**	**−0.4**	← Depreciation

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Relationship of cash flow (CF) and net income (NI)?
When spending on fixed assets, CF = NI − spending on fixed assets
When recording depreciation, CF = NI + depreciation

How Do Net Income and Cash Flow Differ?

Recall from **Module 2**

- The firm spends $60 million producing AI control panels
- None of them have been sold yet



OpenAI, July 15, 2026 (2:15),
"Introducing the Codex Micro"

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Recall from **Module 2**

- The firm owed suppliers $30 million for electronic parts
- The firm pays down the $30 million



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When firm pays off suppliers, CF = NI + change in accounts payable

Operating Activities

Recall from **Module 2**

Operating activities, cash from running the business day to day

Steps to calculate cash flow from operating activities,

1. **Start** with net income
2. **Add** depreciation
3. **Subtract** change in net working capital (NWC)
   - NWC = inventories + accounts receivable − accounts payable
   - NWC, cash tied up in day-to-day operations, in unsold goods and unpaid bills

Why Subtract Change in Net Working Capital (NWC)?

Recall from **Module 2**

When adding to inventories, CF = NI − change in inventories

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$$CF = NI - \text{change in net working capital (NWC)}$$

Investing Activities

Recall from **Module 2**

Investing activities, cash spent buying fixed assets and cash received from selling them

Investing activities show cash flows from,

1. Capital purchased and sold
   - **Capital**, a firm's fixed assets
   - **Capital expenditure**, the amount spent on buying capital during the period
   - Cash flow is *negative* if firms buy capital

Financing Activities

Recall from **Module 2**

Financing activities, cash raised from lenders and shareholders, and cash returned to them

Recap of Part 0

Recall from **Module 2**

- The statement of cash flows, cash receipts and payments over a period
- Net income $\neq$ cash flow
- The statement has three parts
  - **Operating activities**, cash flow = Net income + Depreciation − Change in NWC
  - **Investing activities**, cash spent buying fixed assets and cash received from selling them
  - **Financing activities**, cash raised from lenders and shareholders, and cash returned to them

Module 7, Part 1

Project Cash Flow

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Firm Cash Flow vs. Project Cash Flow

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- **Operating activities**, the same idea, project net income plus depreciation minus the change in NWC
- **Investing activities**, the project's capital investments, buying its assets at the start
- **Financing activities**, *none*, we do not consider financing activities when evaluating a project on its own

Identify Project Cash Flow

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$$\text{Cash flow from a project} = \begin{cases} \text{Project net income} & \text{Component 1} \\ + \text{ Depreciation} & \text{Component 2} \\ \text{Cash flow from changes in NWC} & \text{Component 3} \\ \text{Cash flow from capital investments} & \text{Component 4} \end{cases}$$

Component 1, Project Net Income

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Revenues from the project
– COGS and SG&A expenses for the project
– Depreciation for the project's capital investment
Project EBIT
– Taxes (tax rate $\times$ project EBIT)
Project net income

Component 2, Depreciation

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- To get cash flow from the project, add back depreciation, as in **Module 2**

Example, Project Net Income and Depreciation

- A project requires an investment in fixed assets of $2,000. The project's revenues minus its COGS and SG&A expenses are $1,100 in year 1 and $1,300 in year 2. The investment is depreciated at $1,000 per year. The corporate tax rate is 21%

Example, Project Net Income and Depreciation

- A project requires an investment in fixed assets of $2,000. The project's revenues minus its COGS and SG&A expenses are $1,100 in year 1 and $1,300 in year 2. The investment is depreciated at $1,000 per year. The corporate tax rate is 21%
- Calculate the project's net income plus depreciation for year 1

Example, Project Net Income and Depreciation

- Solution,

Item	$
Revenues − COGS − SG&A expenses	1,100
− Depreciation	1,000
Project EBIT	**100**
− Taxes	21
Project net income	**79**
+ Depreciation	1,000
Project net income + Depreciation	**1,079**

Example, Project Net Income and Depreciation

- Solution,

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- Taxes, $21\% \times$ project EBIT $= 21\% \times 100 = 21$
- Project net income + Depreciation $= 79 + 1{,}000 = \$1{,}079$

Component 3, Cash Flow from Changes in NWC

- Recall from **Module 2**,
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- An increase in the project's NWC is a negative cash flow, and a decrease is a positive cash flow,

Cash flow from changes in NWC = −Change in the project's NWC

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Year	0	1	2	3
The project's NWC	100	100	100	0
Change in NWC	$+100$	0	0	-100
Cash flow from changes in NWC	-100	0	0	$+100$

Component 3, Cash Flow from Changes in NWC

- E.g., a 3-year project needs \$100 of NWC while it operates, recovered when the project ends

Year	0	1	2	3
The project's NWC	100	100	100	0
Change in NWC	$+100$	0	0	-100
Cash flow from changes in NWC	-100	0	0	$+100$

- The \$100 goes out at $t = 0$ and returns only at $t = 3$, so tying up the cash still costs the time value of money

Component 4, Cash Flow from Capital Investments

- If the project requires its own investment in fixed assets to get started (e.g., building a data center), that is a negative cash flow

Component 4, Cash Flow from Capital Investments

- If the project requires its own investment in fixed assets to get started (e.g., building a data center), that is a negative cash flow
- Fixed assets are usually depreciated "straight-line" over time,
  - **Expected life**, the number of years the project uses the fixed assets

$$\text{Annual depreciation} = \frac{\text{Capital investment}}{\text{Expected life}}$$

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Recap of Part 1

- Project net income comes from the project's own income statement, revenues less COGS and SG&A expenses, depreciation, and taxes
- Depreciation is not a cash flow, add it back to the project's net income
- An increase in the project's NWC is a negative cash flow, and a decrease is a positive cash flow
- The investment in fixed assets is a negative cash flow, and the fixed assets are depreciated straight-line over the expected life

Module 7, Part 2

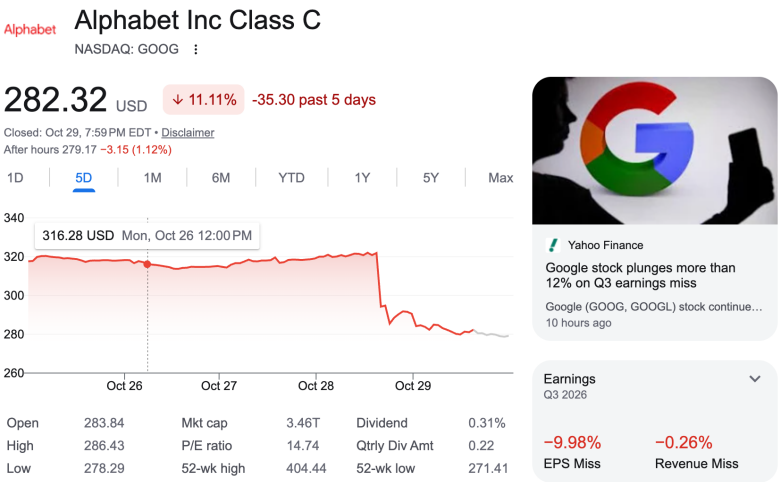
Putting It All Together

Motivating Example, Google's AI Spend under Scrutiny



CNBC, October 28, 2026 (3:06), "Google's AI spend under scrutiny, here's what to know"

Motivating Example, Google's AI Spend under Scrutiny



Example, Putting It All Together

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- Renting out its computing capacity brings in \$240 million a year after COGS and SG&A expenses (electricity, staff, and so on)
- The corporate tax rate is 21%, and the discount rate is 8%
- Should the company build the data center or not?

Identify Project Cash Flow

- Cash flow from a project has four components,

$$\text{Cash flow from a project} = \begin{cases} \text{Project net income} & \text{Component 1} \\ + \text{ Depreciation} & \text{Component 2} \\ \text{Cash flow from changes in NWC} & \text{Component 3} \\ \text{Cash flow from capital investments} & \text{Component 4} \end{cases}$$

Example, Component 1, Project Net Income

- Build the project's income statement, the same in years 1 to 5 ($ millions),

Year	0	1	2	3	4	5

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Year	0	1	2	3	4	5
Revenues − COGS − SG&A expenses		240	240	240	240	240

Example, Component 1, Project Net Income

- Build the project's income statement, the same in years 1 to 5 ($ millions),

Year	0	1	2	3	4	5
Revenues − COGS − SG&A expenses		240	240	240	240	240
− Depreciation		200	200	200	200	200

- Depreciation $= 1{,}000/5 = 200$ a year

Example, Component 1, Project Net Income

- Build the project's income statement, the same in years 1 to 5 ($ millions),

Year	0	1	2	3	4	5
Revenues − COGS − SG&A expenses		240	240	240	240	240
− Depreciation		200	200	200	200	200
Project EBIT		**40**	**40**	**40**	**40**	**40**

- Depreciation $= 1{,}000/5 = 200$ a year

Example, Component 1, Project Net Income

- Build the project's income statement, the same in years 1 to 5 ($ millions),

Year	0	1	2	3	4	5
Revenues − COGS − SG&A expenses		240	240	240	240	240
− Depreciation		200	200	200	200	200
Project EBIT		**40**	**40**	**40**	**40**	**40**
− Taxes		8.4	8.4	8.4	8.4	8.4

- Depreciation $= 1{,}000/5 = 200$ a year
- Taxes $= 21\% \times 40 = 8.4$

Example, Component 1, Project Net Income

- Build the project's income statement, the same in years 1 to 5 ($ millions),

Year	0	1	2	3	4	5
Revenues − COGS − SG&A expenses		240	240	240	240	240
− Depreciation		200	200	200	200	200
Project EBIT		**40**	**40**	**40**	**40**	**40**
− Taxes		8.4	8.4	8.4	8.4	8.4
Project net income		**31.6**	**31.6**	**31.6**	**31.6**	**31.6**

- Depreciation $= 1{,}000/5 = 200$ a year
- Taxes $= 21\% \times 40 = 8.4$

Example, Component 2, Depreciation

- Depreciation is not a cash flow, add it back ($ millions),

Year	0	1	2	3	4	5

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Project net income		31.6	31.6	31.6	31.6	31.6

Example, Component 2, Depreciation

- Depreciation is not a cash flow, add it back ($ millions),

Year	0	1	2	3	4	5
Project net income		31.6	31.6	31.6	31.6	31.6
+ Depreciation		200	200	200	200	200

Example, Component 2, Depreciation

- Depreciation is not a cash flow, add it back ($ millions),

Year	0	1	2	3	4	5
Project net income		31.6	31.6	31.6	31.6	31.6
+ Depreciation		200	200	200	200	200
Project net income + Depreciation		**231.6**	**231.6**	**231.6**	**231.6**	**231.6**

- The cash for the data center went out at $t = 0$, it is counted in Component 4

Example, Component 3, Cash Flow from Changes in NWC

- The project's NWC is 100 while it operates, recovered at $t = 5$ ($ millions),

Year	0	1	2	3	4	5

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- The project's NWC is 100 while it operates, recovered at $t = 5$ ($ millions),

Year	0	1	2	3	4	5
The project's NWC	100	100	100	100	100	0

Example, Component 3, Cash Flow from Changes in NWC

- The project's NWC is 100 while it operates, recovered at $t = 5$ ($ millions),

Year	0	1	2	3	4	5
The project's NWC	100	100	100	100	100	0
Change in NWC	+100	0	0	0	0	−100

Example, Component 3, Cash Flow from Changes in NWC

- The project's NWC is 100 while it operates, recovered at $t = 5$ ($ millions),

Year	0	1	2	3	4	5
The project's NWC	100	100	100	100	100	0
Change in NWC	+100	0	0	0	0	−100
Cash flow from changes in NWC	**−100**	**0**	**0**	**0**	**0**	**+100**

- An increase in NWC is a negative cash flow, and a decrease is a positive cash flow

Example, Component 4, Cash Flow from Capital Investments

- Investment in fixed assets at $t = 0$ ($ millions),

Year	0	1	2	3	4	5

Example, Component 4, Cash Flow from Capital Investments

- Investment in fixed assets at $t = 0$ ($ millions),

Year	0	1	2	3	4	5
Investment in fixed assets	−1,000	0	0	0	0	0

Example, Component 4, Cash Flow from Capital Investments

- Investment in fixed assets at $t = 0$ ($ millions),

Year	0	1	2	3	4	5
Investment in fixed assets	−1,000	0	0	0	0	0
Cash flow from capital investments	**−1,000**	**0**	**0**	**0**	**0**	**0**

Example, Total Cash Flows

- Add up the four components year by year ($ millions),

Year	0	1	2	3	4	5

Example, Total Cash Flows

- Add up the four components year by year ($ millions),

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Project net income + Depreciation	0	231.6	231.6	231.6	231.6	231.6

Example, Total Cash Flows

- Add up the four components year by year ($ millions),

Year	0	1	2	3	4	5
Project net income + Depreciation	0	231.6	231.6	231.6	231.6	231.6
Cash flow from changes in NWC	−100	0	0	0	0	+100

Example, Total Cash Flows

- Add up the four components year by year ($ millions),

Year	0	1	2	3	4	5
Project net income + Depreciation	0	231.6	231.6	231.6	231.6	231.6
Cash flow from changes in NWC	−100	0	0	0	0	+100
Cash flow from capital investments	−1,000	0	0	0	0	0

Example, Total Cash Flows

- Add up the four components year by year ($ millions),

Year	0	1	2	3	4	5
Project net income + Depreciation	0	231.6	231.6	231.6	231.6	231.6
Cash flow from changes in NWC	−100	0	0	0	0	+100
Cash flow from capital investments	−1,000	0	0	0	0	0
Total cash flows	**−1,100**	**231.6**	**231.6**	**231.6**	**231.6**	**331.6**

- At $t = 0$, $-1{,}000 - 100 = -1{,}100$. At $t = 5$, $231.6 + 100 = 331.6$

Example, NPV and the Decision

- Discount the total cash flows at the 8% discount rate ($ millions),

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$$\text{NPV} = -1{,}100 + \frac{231.6}{1.08} + \frac{231.6}{1.08^2} + \frac{231.6}{1.08^3} + \frac{231.6}{1.08^4} + \frac{331.6}{1.08^5} = -107.23$$

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- NPV < 0, so the company should not build the data center

Example, NPV with the Cash Flow Worksheet

One cash flow at a time

Enter	Keys
	CF, **2nd**, **CLR Work**
−1100	**ENTER**, ↓
231.6	**ENTER**, ↓ ↓
231.6	**ENTER**, ↓ ↓
231.6	**ENTER**, ↓ ↓
231.6	**ENTER**, ↓ ↓
331.6	**ENTER**
	NPV (the calculator asks for I)
8	**ENTER**, ↓
	CPT
−107.23	

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231.6	ENTER, ↓ ↓
331.6	ENTER
	NPV (the calculator asks for I)
8	ENTER, ↓
	CPT
−107.23	

Repeated cash flow with frequency

Enter	Keys
	CF, 2nd, CLR Work
−1100	ENTER, ↓
231.6	ENTER, ↓
4	ENTER (the frequency F01), ↓
331.6	ENTER
	NPV (the calculator asks for I)
8	ENTER, ↓
	CPT
−107.23	

Example, IRR with the Cash Flow Worksheet

- Recall from **Module 6**, the IRR is the discount rate at which NPV $= 0$

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- Recall from **Module 6**, the IRR is the discount rate at which $\text{NPV} = 0$
- The same cash flow worksheet, then **IRR**, **CPT**

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	CF, **2nd**, **CLR Work**
-1100	**ENTER**, ↓
231.6	**ENTER**, ↓
4	**ENTER** (the frequency F01), ↓
331.6	**ENTER**
	IRR (shows 0)
	CPT
4.41%	

Example, IRR with the Cash Flow Worksheet

- Recall from **Module 6**, the IRR is the discount rate at which NPV $= 0$
- The same cash flow worksheet, then **IRR**, **CPT**
- IRR (4.41%) $<$ discount rate (8%), reject the project, the same answer as the NPV rule

Enter	Keys
	CF, **2nd**, **CLR Work**
-1100	**ENTER**, ↓
231.6	**ENTER**, ↓
4	**ENTER** (the frequency F01), ↓
331.6	**ENTER**
	IRR (shows 0)
	CPT
4.41%	

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1. Build the project's income statement year by year, then add back depreciation, project net income + depreciation (Components 1 and 2)

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How to Attack a Full-Scale Problem

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1. Build the project's income statement year by year, then add back depreciation, project net income + depreciation (Components 1 and 2)
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3. Add the cash flow from capital investments, the investment in fixed assets at the start and any terminal cash flows at the end (Component 4)
4. Total the cash flows year by year, discount at the discount rate, and apply the NPV rule

Module 7, Part 3

The Principles behind Project Cash Flows

General Principles for Identifying Project Cash Flow

- Look back at the data center table. It quietly followed three principles,

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1. Discount cash flows, not net income
2. Exclude cash flows from financing activities
3. Consider all and only incremental cash flows

Principle 1, Discount Cash Flows, Not Net Income

- Net income $\neq$ cash flows

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- Differences in how capital investments, depreciation, and NWC are handled
- We always discount the actual cash flows to their present value
- We do not discount net income

Example, Net Income vs. Cash Flows

- A project requires an investment in fixed assets of $2,000 and lasts 2 years. The fixed assets are depreciated straight-line at $1,000 per year

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- The discount rate is 10%

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- Compare the NPV using net income to the NPV using cash flows

Clearly a money-loser!

Example, Net Income vs. Cash Flows

- Solution,

Year	0	1	2

Example, Net Income vs. Cash Flows

- Solution,

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Revenues − COGS − SG&A expenses		1,500	500
− Depreciation		1,000	1,000

Example, Net Income vs. Cash Flows

- Solution,

Year	0	1	2
Revenues − COGS − SG&A expenses		1,500	500
− Depreciation		1,000	1,000
Project net income		**500**	**−500**

Example, Net Income vs. Cash Flows

- Solution,

Year	0	1	2
Revenues − COGS − SG&A expenses		1,500	500
− Depreciation		1,000	1,000
Project net income		**500**	**−500**
+ Depreciation		1,000	1,000
Project net income + Depreciation	**0**	**1,500**	**500**

Example, Net Income vs. Cash Flows

- Solution,

Year	0	1	2
Revenues − COGS − SG&A expenses		1,500	500
− Depreciation		1,000	1,000
Project net income		**500**	**−500**
+ Depreciation		1,000	1,000
Project net income + Depreciation	**0**	**1,500**	**500**
Cash flow from capital investments	−2,000	0	0

Example, Net Income vs. Cash Flows

- Solution,

Year	0	1	2
Revenues − COGS − SG&A expenses		1,500	500
− Depreciation		1,000	1,000
Project net income		**500**	**−500**
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Cash flow from capital investments	−2,000	0	0
Total cash flows	**−2,000**	**1,500**	**500**

Example, Net Income vs. Cash Flows

- Solution,

Year	0	1	2
Revenues − COGS − SG&A expenses		1,500	500
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Project net income		**500**	**−500**
+ Depreciation		1,000	1,000
Project net income + Depreciation	**0**	**1,500**	**500**
Cash flow from capital investments	−2,000	0	0
Total cash flows	**−2,000**	**1,500**	**500**

- NPV using net income $= \frac{500}{1.10} - \frac{500}{1.10^2} = \$41.32$, which says take the project

Example, Net Income vs. Cash Flows

- Solution,

Year	0	1	2
Revenues − COGS − SG&A expenses		1,500	500
− Depreciation		1,000	1,000
Project net income		**500**	**−500**
+ Depreciation		1,000	1,000
Project net income + Depreciation	**0**	**1,500**	**500**
Cash flow from capital investments	−2,000	0	0
Total cash flows	**−2,000**	**1,500**	**500**

- NPV using net income $= \frac{500}{1.10} - \frac{500}{1.10^2} = \$41.32$, which says take the project
- NPV using cash flows $= -2{,}000 + \frac{1{,}500}{1.10} + \frac{500}{1.10^2} = -\$223.14$

Example, Net Income vs. Cash Flows

- Solution,

Year	0	1	2
Revenues − COGS − SG&A expenses		1,500	500
− Depreciation		1,000	1,000
Project net income		**500**	**−500**
+ Depreciation		1,000	1,000
Project net income + Depreciation	**0**	**1,500**	**500**
Cash flow from capital investments	−2,000	0	0
Total cash flows	**−2,000**	**1,500**	**500**

- NPV using net income $= \frac{500}{1.10} - \frac{500}{1.10^2} = \$41.32$, which says take the project
- NPV using cash flows $= -2{,}000 + \frac{1{,}500}{1.10} + \frac{500}{1.10^2} = -\$223.14$
- NPV < 0, so the project should be rejected, net income gets it wrong

Principle 2, Exclude Cash Flows from Financing Activities

- Project cash flows exclude cash flows from financing activities

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- Project cash flows exclude cash flows from financing activities
- The mini-income statement for this data center project has no interest expense row, even if the company borrows to build it

Principle 3, Consider All and Only Incremental Cash Flows

- We need to discount **incremental** cash flows
- What does "incremental" mean in capital budgeting?

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- Ask yourself, "would the cash flow still exist if the project did not exist?"
  - If yes, do *not* include it in the analysis
  - If no, include it

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- We need to discount **incremental** cash flows
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Incremental cash flow = Cash flow with the project − Cash flow without it

Incremental Cash Flows, a Few Rules That Are Easy to Forget

a. Include all indirect effects

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Incremental Cash Flows, a Few Rules That Are Easy to Forget

a. Include all indirect effects
b. Exclude sunk costs
c. Include opportunity costs
d. Include cash flows from changes in NWC
e. Include terminal cash flows

Rule a, Indirect Effects

- **Indirect effects**, the effects of a project on the cash flows of the company's other businesses

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- **Indirect effects**, the effects of a project on the cash flows of the company's other businesses
- E.g., **cannibalization**, a new product cutting into the sales of the company's existing products

Rule a, Indirect Effects

- Nintendo plans a new-generation gaming console
- It requires an investment of $200m and produces cash flows of $400m and $800m in years 1 and 2





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- But the launch will hurt sales of the older console
- Without the new console, the old one is forecast to create $800m and $1,000m in years 1 and 2, with the new console, only $500m and $400m





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- But the launch will hurt sales of the older console
- Without the new console, the old one is forecast to create $800m and $1,000m in years 1 and 2, with the new console, only $500m and $400m
- What is the **incremental** cash flow of the new console?





Rule a, Indirect Effects

- Solution ($ millions),

Year	0	1	2
Cash flow from investment in and sales of the new one	−200	400	800
Cash flow from the old console without the new one		800	1,000
Cash flow from the old console with the new one		500	400
Change in cash flow from the old console		−300	−600
Incremental cash flow of the new console	−200	$400 - 300 = 100$	$800 - 600 = 200$

Rule a, Indirect Effects

- Solution ($ millions),

Year	0	1	2	
Cash flow from investment in and sales of the new one	−200	400	800	← Direct effect
Cash flow from the old console without the new one		800	1,000	
Cash flow from the old console with the new one		500	400	
Change in cash flow from the old console		−300	−600	← Indirect effect
Incremental cash flow of the new console	−200	$400 - 300 = 100$	$800 - 600 = 200$	

Rule b, Sunk Costs

- **Sunk costs**, expenses that have already been paid, or expenses that can no longer be changed
  - E.g., previous R&D (already paid), rent under an unbreakable lease (can no longer be changed)



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- **Sunk costs**, expenses that have already been paid, or expenses that can no longer be changed
  - E.g., previous R&D (already paid), rent under an unbreakable lease (can no longer be changed)
- A sunk cost is the same with or without the project $\rightarrow$ it is not incremental and should not affect the NPV $\rightarrow$ Exclude it!
- Many sunk costs are "related" to the project. But if the decision does not change them, they are not incremental



Example, Sunk Costs

- Your company has developed a new product that is almost ready to launch. The R&D cost $0.6 million and was completed last year
- Launching requires another $1 million investment (marketing and so on), with a present value of the project's future cash flows of $1.5 million

Example, Sunk Costs

- Your company has developed a new product that is almost ready to launch. The R&D cost $0.6 million and was completed last year
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- Your company has developed a new product that is almost ready to launch. The R&D cost $0.6 million and was completed last year
- Launching requires another $1 million investment (marketing and so on), with a present value of the project's future cash flows of $1.5 million
- Proceed with the project or abort?
  - The $0.6 million R&D is a sunk cost, the same whether you proceed or abort $\rightarrow$ Exclude it!

Example, Sunk Costs

- Your company has developed a new product that is almost ready to launch. The R&D cost \$0.6 million and was completed last year
- Launching requires another \$1 million investment (marketing and so on), with a present value of the project's future cash flows of \$1.5 million
- Proceed with the project or abort?
  - The \$0.6 million R&D is a sunk cost, the same whether you proceed or abort $\rightarrow$ Exclude it!
  - NPV $= \$1.5 - \$1 = \$0.5$ million $> 0 \rightarrow$ Proceed!

Rule c, Opportunity Costs

- A resource is almost never “free”
  - E.g., a piece of land the company already owns is not free, the firm could use it for another project, rent it out, or sell it

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- What is the actual cost of building the factory?

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  - The land is not free, its opportunity cost is the $3m you give up by not selling it

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- What is the actual cost of building the factory?
  - The land is not free, its opportunity cost is the \$3m you give up by not selling it
  - Actual cost = \$10m + \$3m = \$13m

Rule d, Changes in NWC

- If a project needs its own NWC, the cash flows from changes in NWC belong in the analysis

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- If a project needs its own NWC, the cash flows from changes in NWC belong in the analysis
- The project's NWC often keeps changing during the project, and is often recovered at the end

Practice Problem

- A company's project runs for 3 years. In $ thousands, its revenues are 1,000 in year 1, 2,000 in year 2, and 1,000 in year 3
- The project's NWC at each date must be 10% of the following one-year period's revenues

Practice Problem

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- What are the cash flows from changes in NWC?

Practice Problem

- Solution ($ thousands), NWC at each date is 10% of the following one-year period's revenues,

Year	0	1	2	3
Revenues		1,000	2,000	1,000
The project's NWC	100	200	100	0
Change in NWC	+100	+100	−100	−100
Cash flow from changes in NWC	−100	−100	+100	+100

Practice Problem

- Solution ($ thousands), NWC at each date is 10% of the following one-year period's revenues,

Year	0	1	2	3
Revenues		1,000	2,000	1,000
The project's NWC	100	200	100	0
Change in NWC	+100	+100	−100	−100
Cash flow from changes in NWC	−100	−100	+100	+100

- NWC follows revenues up and down, and is fully recovered by the end of the project

Rule e, Terminal Cash Flows

- At the end, there may be cash flows from shutting down the project, the "terminal" cash flows



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  - E.g., oil wells require "capping" when they no longer produce



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- They can also be negative
  - E.g., oil wells require "capping" when they no longer produce
  - But apparently many companies still do this, Bloomberg, "It Pays to Not Pay Your Debts"



Practice Problem

- A company plans a new manufacturing plant on land it already owns, but the existing buildings must be demolished. Which of the following are incremental cash flows?

a. Money already spent on engineering design of the new plant
b. The cost of a new access road put in last year
c. The market value of the land
d. The market value of the existing buildings
e. Demolition and clearance costs
f. Construction cost of the new plant
g. The initial investment in inventories of raw materials
h. Lost cash flows on other projects due to executive time spent on the new plant

Practice Problem

- A company plans a new manufacturing plant on land it already owns, but the existing buildings must be demolished. Which of the following are incremental cash flows?

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Recap of Part 3

- Discount cash flows, not net income, and exclude cash flows from financing activities

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- Discount cash flows, not net income, and exclude cash flows from financing activities
- Consider all and only incremental cash flows, "would the cash flow still exist if the project did not exist?"
- Include indirect effects, opportunity costs, changes in NWC, and terminal cash flows, exclude sunk costs

Module 7, Part 4

Scenario Analysis

Example, the Data Center Again

- Recall the data center from Part 2 ($ millions),
  - An investment in fixed assets of 1,000 that lasts 5 years, depreciated straight-line over 5 years

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  - An investment in fixed assets of 1,000 that lasts 5 years, depreciated straight-line over 5 years
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  - It brings in 240 a year after COGS and SG&A expenses

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  - It brings in 240 a year after COGS and SG&A expenses
  - The corporate tax rate is 21%, and the discount rate is 8%

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  - The corporate tax rate is 21%, and the discount rate is 8%
- This time, build the whole table in one go

Example, the Data Center Again

- Solution ($ millions), Components 1 and 2, then 3, then 4, then the totals

Year	0	1	2	3	4	5

Example, the Data Center Again

- Solution ($ millions), Components 1 and 2, then 3, then 4, then the totals

Year	0	1	2	3	4	5
Revenues − COGS − SG&A expenses		240	240	240	240	240

Example, the Data Center Again

- Solution ($ millions), Components 1 and 2, then 3, then 4, then the totals

Year	0	1	2	3	4	5
Revenues − COGS − SG&A expenses		240	240	240	240	240
− Depreciation		200	200	200	200	200

Example, the Data Center Again

- Solution ($ millions), Components 1 and 2, then 3, then 4, then the totals

Year	0	1	2	3	4	5
Revenues − COGS − SG&A expenses		240	240	240	240	240
− Depreciation		200	200	200	200	200
Project EBIT		**40**	**40**	**40**	**40**	**40**

Example, the Data Center Again

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Year	0	1	2	3	4	5
Revenues − COGS − SG&A expenses		240	240	240	240	240
− Depreciation		200	200	200	200	200
Project EBIT		**40**	**40**	**40**	**40**	**40**
− Taxes		8.4	8.4	8.4	8.4	8.4

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Year	0	1	2	3	4	5
Revenues − COGS − SG&A expenses		240	240	240	240	240
− Depreciation		200	200	200	200	200
Project EBIT		**40**	**40**	**40**	**40**	**40**
− Taxes		8.4	8.4	8.4	8.4	8.4
Project net income		**31.6**	**31.6**	**31.6**	**31.6**	**31.6**

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Project net income		**31.6**	**31.6**	**31.6**	**31.6**	**31.6**
+ Depreciation		200	200	200	200	200

Example, the Data Center Again

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− Taxes		8.4	8.4	8.4	8.4	8.4
Project net income		**31.6**	**31.6**	**31.6**	**31.6**	**31.6**
+ Depreciation		200	200	200	200	200
Project net income + Depreciation		**231.6**	**231.6**	**231.6**	**231.6**	**231.6**

Example, the Data Center Again

- Solution ($ millions), Components 1 and 2, then 3, then 4, then the totals

Year	0	1	2	3	4	5
Revenues − COGS − SG&A expenses		240	240	240	240	240
− Depreciation		200	200	200	200	200
Project EBIT		**40**	**40**	**40**	**40**	**40**
− Taxes		8.4	8.4	8.4	8.4	8.4
Project net income		**31.6**	**31.6**	**31.6**	**31.6**	**31.6**
+ Depreciation		200	200	200	200	200
Project net income + Depreciation		**231.6**	**231.6**	**231.6**	**231.6**	**231.6**
Cash flow from changes in NWC	−100	0	0	0	0	+100

Example, the Data Center Again

- Solution ($ millions), Components 1 and 2, then 3, then 4, then the totals

Year	0	1	2	3	4	5
Revenues − COGS − SG&A expenses		240	240	240	240	240
− Depreciation		200	200	200	200	200
Project EBIT		**40**	**40**	**40**	**40**	**40**
− Taxes		8.4	8.4	8.4	8.4	8.4
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+ Depreciation		200	200	200	200	200
Project net income + Depreciation		**231.6**	**231.6**	**231.6**	**231.6**	**231.6**
Cash flow from changes in NWC	−100	0	0	0	0	+100
Cash flow from capital investments	−1,000	0	0	0	0	0

Example, the Data Center Again

- Solution ($ millions), Components 1 and 2, then 3, then 4, then the totals

Year	0	1	2	3	4	5
Revenues − COGS − SG&A expenses		240	240	240	240	240
− Depreciation		200	200	200	200	200
Project EBIT		**40**	**40**	**40**	**40**	**40**
− Taxes		8.4	8.4	8.4	8.4	8.4
Project net income		**31.6**	**31.6**	**31.6**	**31.6**	**31.6**
+ Depreciation		200	200	200	200	200
Project net income + Depreciation		**231.6**	**231.6**	**231.6**	**231.6**	**231.6**
Cash flow from changes in NWC	−100	0	0	0	0	+100
Cash flow from capital investments	−1,000	0	0	0	0	0
Total cash flows	**−1,100**	**231.6**	**231.6**	**231.6**	**231.6**	**331.6**

Example, the Data Center Again

- Solution ($ millions), Components 1 and 2, then 3, then 4, then the totals

Year	0	1	2	3	4	5
Revenues − COGS − SG&A expenses		240	240	240	240	240
− Depreciation		200	200	200	200	200
Project EBIT		**40**	**40**	**40**	**40**	**40**
− Taxes		8.4	8.4	8.4	8.4	8.4
Project net income		**31.6**	**31.6**	**31.6**	**31.6**	**31.6**
+ Depreciation		200	200	200	200	200
Project net income + Depreciation		**231.6**	**231.6**	**231.6**	**231.6**	**231.6**
Cash flow from changes in NWC	−100	0	0	0	0	+100
Cash flow from capital investments	−1,000	0	0	0	0	0
Total cash flows	**−1,100**	**231.6**	**231.6**	**231.6**	**231.6**	**331.6**
NPV	**−107.23**					

- NPV = −$107.23 million, reject the project

Scenario Analysis

- The inputs behind a project's cash flows are forecasts, and forecasts can be wrong

Scenario Analysis

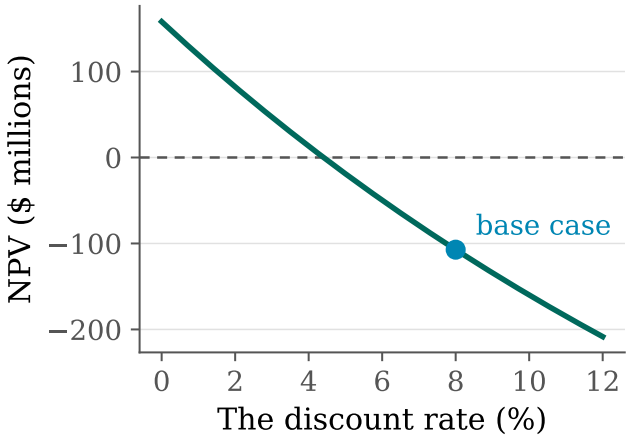
- The inputs behind a project's cash flows are forecasts, and forecasts can be wrong
- **Scenario analysis**, recompute the NPV under different assumptions about the inputs

Scenario Analysis

- The inputs behind a project's cash flows are forecasts, and forecasts can be wrong
- **Scenario analysis**, recompute the NPV under different assumptions about the inputs
- E.g., for the data center, how does the NPV change with the discount rate, with what it brings in a year, or with the investment in fixed assets?

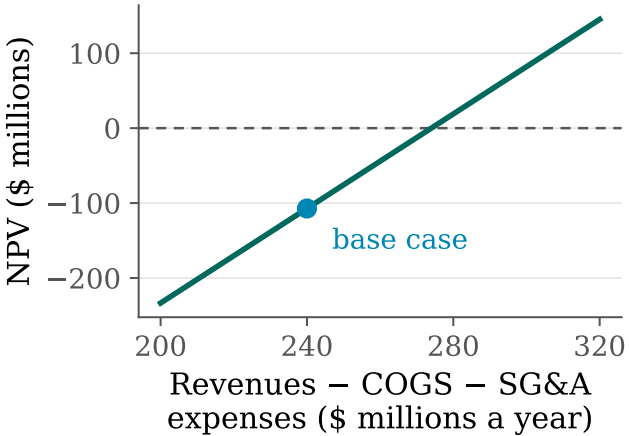
Example, Scenario Analysis

- NPV falls as the discount rate rises, positive only below the IRR of 4.41%



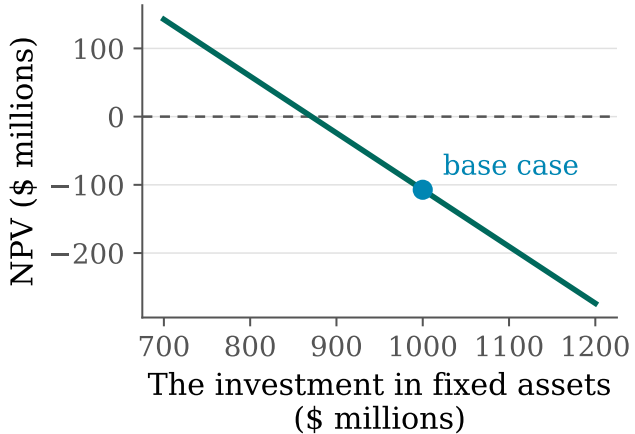
Example, Scenario Analysis

- The data center must bring in about 274 a year after COGS and SG&A expenses to break even



Example, Scenario Analysis

- At 240 a year, the investment in fixed assets must come in under about 871 to break even



Using AI for Scenario Analysis

- Scenario analysis recomputes the same NPV many times, a perfect job for AI

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- In-class demo, I give the AI a prompt and run the code it returns

Module 7, Part 5

Practice Problem

Practice Problem

- A company makes bottles and just developed a new bottle design
- The new design needs an investment in fixed assets (machinery) of $6 million, depreciated straight-line over 6 years

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- A company makes bottles and just developed a new bottle design
- The new design needs an investment in fixed assets (machinery) of \$6 million, depreciated straight-line over 6 years
- It needs an increase in NWC of \$250,000 at $t = 0$, recovered fully at $t = 6$

Practice Problem

- A company makes bottles and just developed a new bottle design
- The new design needs an investment in fixed assets (machinery) of \$6 million, depreciated straight-line over 6 years
- It needs an increase in NWC of \$250,000 at $t = 0$, recovered fully at $t = 6$
- It sells 1 million bottles in year 1, growing at 5% a year over the six years. Each bottle sells for \$4.00, and its COGS is \$2.00. There are no SG&A expenses

Practice Problem

- A company makes bottles and just developed a new bottle design
- The new design needs an investment in fixed assets (machinery) of \$6 million, depreciated straight-line over 6 years
- It needs an increase in NWC of \$250,000 at $t = 0$, recovered fully at $t = 6$
- It sells 1 million bottles in year 1, growing at 5% a year over the six years. Each bottle sells for \$4.00, and its COGS is \$2.00. There are no SG&A expenses
- The project ends after 6 years. The corporate tax rate is 21%, and the discount rate is 12%

Practice Problem

- A company makes bottles and just developed a new bottle design
- The new design needs an investment in fixed assets (machinery) of \$6 million, depreciated straight-line over 6 years
- It needs an increase in NWC of \$250,000 at $t = 0$, recovered fully at $t = 6$
- It sells 1 million bottles in year 1, growing at 5% a year over the six years. Each bottle sells for \$4.00, and its COGS is \$2.00. There are no SG&A expenses
- The project ends after 6 years. The corporate tax rate is 21%, and the discount rate is 12%
- What is the project's NPV?

Solution, Component 1, Project Net Income

- Build the project's income statement year by year ($ thousands),

Year	0	1	2	3	4	5	6

Solution, Component 1, Project Net Income

- Build the project's income statement year by year ($ thousands),

Year	0	1	2	3	4	5	6
Bottles sold (thousands)		1,000	1,050	1,103	1,158	1,216	1,276

Solution, Component 1, Project Net Income

- Build the project's income statement year by year ($ thousands),

Year	0	1	2	3	4	5	6
Bottles sold (thousands)		1,000	1,050	1,103	1,158	1,216	1,276
Revenues		4,000	4,200	4,410	4,631	4,862	5,105

- Revenues $= \$4.00\times$ bottles sold, which grow 5% a year

Solution, Component 1, Project Net Income

- Build the project's income statement year by year ($ thousands),

Year	0	1	2	3	4	5	6
Bottles sold (thousands)		1,000	1,050	1,103	1,158	1,216	1,276
Revenues		4,000	4,200	4,410	4,631	4,862	5,105
− COGS		2,000	2,100	2,205	2,315	2,431	2,553

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+ Depreciation		1,000	1,000	1,000	1,000	1,000	1,000

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Project net income + Depreciation		**1,790**	**1,869**	**1,952**	**2,039**	**2,130**	**2,227**

- The cash for the machinery went out at $t = 0$, it is counted in Component 4

Solution, Component 3, Cash Flow from Changes in NWC

- The project's NWC is 250 while it operates, recovered at $t = 6$ ($ thousands),

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The project's NWC	250	250	250	250	250	250	0

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The project's NWC	250	250	250	250	250	250	0
Change in NWC	+250	0	0	0	0	0	−250

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The project's NWC	250	250	250	250	250	250	0
Change in NWC	+250	0	0	0	0	0	−250
Cash flow from changes in NWC	**−250**	**0**	**0**	**0**	**0**	**0**	**+250**

- An increase in NWC is a negative cash flow, and a decrease is a positive cash flow

Solution, Component 4, Cash Flow from Capital Investments

- Investment in fixed assets (the machinery) at $t = 0$ ($ thousands),

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Year	0	1	2	3	4	5	6
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Year	0	1	2	3	4	5	6
Investment in fixed assets	−6,000	0	0	0	0	0	0
Cash flow from capital investments	**−6,000**	**0**	**0**	**0**	**0**	**0**	**0**

Solution, Total Cash Flows

- Add up the four components year by year ($ thousands),

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Cash flow from changes in NWC	−250	0	0	0	0	0	+250
Cash flow from capital investments	−6,000	0	0	0	0	0	0
Total cash flows	**−6,250**	**1,790**	**1,869**	**1,952**	**2,039**	**2,130**	**2,477**

- At $t = 0$, $-6{,}000 - 250 = -6{,}250$. At $t = 6$, $2{,}227 + 250 = 2{,}477$

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- NPV > 0, so the company should launch the new design

Key Takeaways

- Cash flow from a project adds project net income, depreciation, cash flow from changes in NWC, and cash flow from capital investments

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- Discount cash flows, not net income, and exclude cash flows from financing activities
- Only incremental cash flows count, include indirect effects, opportunity costs, changes in NWC, and terminal cash flows, exclude sunk costs
- Lay the years out in a table, total the cash flows, discount, and apply the NPV rule

Thank you!

Please let me know if you have any questions.